



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E CSE
Course Code & Title : AI3021 & IT in Agricultural System
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV/Weather Prediction Models/Seasonal Climate Forecasting
- **Course Outcome:**4
- **Topic Learning Outcome:** 32,33
- **Activity Chosen:** Reflection
- **Justification:**
 - Seasonal prediction is a highly relevant topic that helps students understand how long-range weather forecasts are generated using scientific models, climate patterns, and atmospheric data.
 - Justifying its use in an innovative teaching reflection is important because seasonal forecasting involves complex concepts such as temperature variation, rainfall prediction, ENSO patterns, and monsoon behavior that students can understand more effectively through active, student-centered methods rather than traditional lectures.
 - By applying innovative strategies like collaborative analysis, data interpretation, simulations, or peer discussion, students actively engage with real climate data and learn how seasonal predictions support agriculture, disaster preparedness, water management, and public planning.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The implementation of the innovative practice on *Seasonal Prediction* was carried out on **10.02.2025**, beginning with an introduction to essential concepts such as temperature variation, monsoon patterns, rainfall behavior, and the influence of global climate systems like El Niño and La Niña.
 - To support effective learning, students were provided with pre-class seasonal forecast data, including rainfall charts, temperature anomaly maps, and simplified climate model outputs. During the session, students were divided into groups, and Ms.Pavitraree and her team were assigned to interpret rainfall prediction datasets.
 - Each group analyzed the provided climate data, identified trends, and interpreted what the indicators suggested for the upcoming season. They discussed the implications of these predictions on agriculture, water management, and disaster preparedness. Selected groups,

including Pavitra sri's team, presented their data-driven findings to the class, explaining the patterns they observed and their reasoning behind the predictions.

- The instructor concluded the session by summarizing key concepts, clarifying doubts, and connecting student interpretations to real-world forecasting models.
- In the final reflection, students shared that working directly with climate data improved their analytical ability, strengthened their understanding of seasonal prediction models, and enhanced their awareness of how such predictions influence real-world decision-making.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO8	PO10	PO12	PSO3
CO4	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO5	PO10	PO12	PSO2
	3	3	3	1	2	2	2
Justification for correlation	Student demonstrating testing practices involves applying basic engineering principles to understand and testing concepts and methodologies	Student carrying out advanced types of testing requires analyzing complex problems and identifying suitable testing techniques to address them effectively.	Student advanced testing practices involve investigating complex issues and applying advanced techniques to uncover underlying defects and vulnerabilities.	Student demonstrating testing practices includes utilizing modern testing tools and technologies to automate testing processes and enhance testing efficiency.	Student effective communication of testing practices and results to stakeholders ensures alignment and support for testing efforts, contributing to successful project outcomes.	Student demonstrating testing practices requires a commitment to continuous learning to stay updated with evolving testing methodologies and technologies.	Students can demonstrate the ability to ensure the reliability and security of IT solutions, aligning with the application of IoT, cloud computing, and cybersecurity knowledge.

- **Images / Screenshot of the practice:**

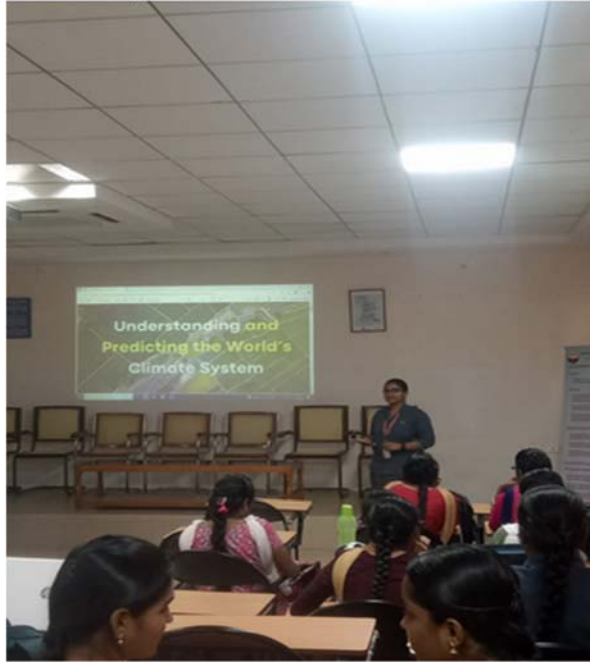


Fig 1: Student discussed the topics

Reflective Critique:

❖ **Feedback from Students and Stakeholders:**

- Students shared that the activity on *Seasonal Forecasting* helped them understand complex climate concepts more clearly and confidently. They appreciated the pre-class materials, which allowed them to review important topics such as temperature anomalies, rainfall prediction, monsoon patterns, ENSO (El Niño–Southern Oscillation), and drought/heatwave indicators at their own pace.
- During the group-based discussion and data interpretation session, students found it easier to clarify doubts and strengthen their understanding by analyzing real climate datasets and seasonal prediction charts. Stakeholders also observed increased engagement, as students were able to connect theoretical climate science concepts with real-world applications in agriculture, water management, and disaster preparedness.

❖ **Benefits of the Practice:**

- Collaborative learning improved communication, critical thinking, and teamwork among students.
- Explaining concepts such as ENSO patterns, monsoon variability, and rainfall forecasting strengthened students' conceptual clarity.
- Hands-on interpretation of seasonal prediction data helped students connect theory with real-world climate scenarios and decision-making.
- Students developed stronger analytical and problem-solving skills by examining trends in climate data and predicting seasonal outcomes.

- The activity increased motivation and awareness of the importance of climate prediction in agriculture, public planning, and environmental sustainability.

❖ **Challenges Faced:**

- Some students were hesitant to participate actively due to the technical depth of climate modeling and prediction.
- A few students struggled with understanding advanced terms such as sea surface temperature anomalies, atmospheric circulation patterns, and pressure indices.
- Students required additional guidance to interpret complex climate charts and graphical seasonal model outputs.
- Limited prior exposure to climate forecasting concepts made some learners dependent on faculty support.
- Ensuring equal participation among all team members was challenging due to differences in confidence and analytical skill levels.

References:

- ❖ https://en.wikipedia.org/wiki/Peer_learning
- ❖ [Peer to Peer Learning - Examples, Benefits & Strategies](#)
- ❖ <https://whatfix.com/blog/peer-to-peer-learning/>

Signature of Faculty Member

HOD